PARADOX: Temporal Programming

Official Language Documentation & Cheat Sheet

Temporal Entities (Custom Data Types)

Paradox Type	Standard	Description	Example
cycle	int	Numeric Data	2100
trace	string	Residual text left in the timeline.	"Anomaly detected"
canon	boolean	Unchangeable truth or falsehood.	fixed (true) erased (false)

Timeline Mechanics (Core Operators & Keywords)

Keyword/Symbol	Standard
<<	= (assignment)
@	; (delimiter)
observe	print
+, -, *, /	Standard

Core Grammar Rules (Syntax)

Action	Structure	Example
Standard	[DATATYPE] [ID] << [LITERAL] @	cycle timeSkip << 2077@
Observation	[OUTPUT_KEYWORD] [ID] @	observe timeSkip@
Chronomath	[DATATYPE] [ID] << [LITERAL/ID] [MATH_OP] [LITERAL] @	cycle offset << 10 + 5@

Advanced Mechanics

Feature / Concept	Exact Grammar Structure	Description

Entity Blueprint	entity [NAME] {
	[DATATYPE] [ID] << [LITERAL]@
	}
		Declares a blueprint for a complex timeline object.

Object Instantiation	[CLASS_ID] [OBJECT_ID] << invoke [CLASS_ID]@	Allocates isolated temporal memory for a new object based on an entity blueprint.
Bootstrap Loop	bootstrap ([ID] [OP] [LITERAL]) { [ID] << [ID] [MATH] [LITERAL]@ } origin@	Traps the timeline in a localized chronological loop until the origin condition fails.
State Snapshot	snapshot [LITERAL]@	Archives an exact replica of the active Universal Symbol Table under a string hash.
State Revert	revert [LITERAL] @	Wipes current memory and restores the Universal Symbol Table to the requested snapshot.
Variable Pruning	Prune [ID]@	Wipes a specific variable from the Universal Symbol Table
Branching	branch@ and prune branch@	Branches timeline into a deeper scope level, while prune branch collapses the current branch to a previous scope level. It is the same as entering a new block ({) and exiting a block (})

The Official Test Case Matrix

Test Type	Input String	Expected Compiler Narrative	Expected Status

TC1: Happy Path	cycle enginePower << 100 @
	snapshot "Baseline" @
	entity Shield {
	canon active << fixed @
	}
	Shield aegis << invoke Shield @
	bootstrap (enginePower > 50) {
	enginePower << enginePower - 25 @
	} origin @
	observe enginePower @
		"Verified all phases, data stored..."	**PASS**

TC2: Syntax Error	cycle velocity 88 @ observe velocity @	"Error: Assignment Operator..."	**FAIL**
TC3: Semantic Error	cycle temperature << "Critical" @ observe temperature @	"Error: Type Mismatch cycle vs trace_literal)..."	**FAIL**
TC4: Lexical Error	canon *checkTimeline << true@	"Error: Foreign artifact “*” ecountered."	**FAIL**

STORY MODE WALKTHROUGH

Mission 1 cycle energy << 50@

Mission 2: trace beacon << "SOS"@

Mission 3: snapshot "Fracture"@

Mission 4: prune energy @

Mission 5: revert "Fracture"@

Mission 6: canon paradox << erased@

Mission 7: entity Shield {

canon active << fixed@

}

Mission 8: Shield aegis << invoke Shield@

Mission 9: observe aegis @

Mission 10: bootstrap (energy > 0) {

energy << energy - 10 @

} origin @